

Eleanor, 84 · stood up from her recliner and fell · unable to bear weight · right leg shortened and externally rotated · pulses intact · **displaced femoral neck fracture**

BIO 004 · WEEK 2 · CASE DEEP DIVE

The Lower Extremity

A grandmother on the floor. A shortened, externally rotated leg. The blood

PATIENT CASE

Eleanor is 84, found on the living room floor by her granddaughter **several hours after a fall**. She had stood up from her recliner and her right leg gave out. She was unable to get herself up and she could not reach her phone.

In the ED she is alert and oriented but in significant pain. On exam the **right leg is visibly shortened compared to the left and externally rotated**; she cannot lift it off the bed. **Distal pulses, sensation, and motor function are intact** in the foot. X-ray of the right hip shows a **displaced fracture of the femoral neck**. The orthopedic surgeon explains to the family that the **blood supply to the femoral head is at risk** and discusses **hemiarthroplasty** (replacement of the femoral head).

Eleanor's injury threads every concept on this page: the pelvic girdle that her trunk sat on, the femur she shortened, the blood supply that may or may not survive the fracture, and the muscles that pull her shortened leg into its classic external rotation. Understanding her hip is the difference between fixing the bone and replacing the head.

GUIDING QUESTION

What bones form the lower extremity, why is the femoral neck the most dangerous place in the lower limb to break, and why is Eleanor's leg shortened and rotated outward when she lies still?

1

OVERVIEW

Four regions, built for weight bearing

The lower extremity has four regions: **pelvic girdle**, **thigh**, **leg** (knee to ankle, NOT the whole leg in anatomical terms), and **foot**. 31 bones per side. Compared to the upper limb, the lower limb sacrifices range of motion for stability and weight bearing. Joints are tighter, sockets are deeper, ligaments are stronger.

The most dangerous bone in the lower limb is the **femoral neck**, not because it breaks most often but because its blood supply is so tenuous that a displaced fracture can kill the head. Eleanor's injury is exactly that.

Three bones that fuse into one, two sides that meet in the middle

Each side of the pelvis is a single bone in the adult (the **os coxae** or hip bone), but it formed from three bones that fused around adolescence:



Ilium

The flaring superior portion. The iliac crest is what you palpate at your waist.



Ischium

Posterior inferior. The ischial tuberosity is what you sit on.



Pubis

Anterior inferior. The two pubic bones meet in the midline at the *pubic symphysis*.



Acetabulum

The deep cup-shaped socket on the lateral pelvis where all three bones meet. Receives the head of the femur. Deeper and more stable than the glenoid of the shoulder.



Obturator foramen

Large opening below the acetabulum. Covered by membrane in life.



Sacroiliac joint

Where the ilium meets the sacrum posteriorly.
Very stable. Transmits body weight from the spine to the pelvis to the legs.

Largest bone in the body, and the neck that worries the surgeon

The **femur** is the bone of the thigh and the largest bone in the body. Proximal to distal:

● Head

Spherical, fits in the acetabulum. Articular surface is a complete ball except for the small *fovea capitis* where the ligament of the head attaches.

● Neck

Narrow segment between head and trochanters. Angled medially. *Eleanor's fracture site*. The most clinically important site of fracture in the elderly.

● Greater trochanter

Lateral prominence at the top of the shaft. Easily palpable. Attachment for gluteus medius and minimus.

● Lesser trochanter

Medial, smaller. Attachment for iliopsoas.

● Shaft

Long and slightly bowed. The strongest tubular bone.

● Distal end

Medial and lateral condyles articulate with the tibia. The patellar surface anteriorly articulates with the kneecap.

The blood supply to the femoral head. The femoral head is fed mainly by the *medial femoral circumflex artery*, whose branches wrap around the neck (retinacular arteries) before entering the head. A small additional supply comes through the ligament of the head from the obturator artery. When the femoral neck fractures and displaces, the retinacular arteries are torn. If too much vessel is lost, the head necroses (avascular necrosis). That is why a displaced femoral neck fracture in an older patient is usually treated by replacing the head (hemiarthroplasty) rather than trying to fix it (open reduction internal fixation).

The largest joint, the bones that support it

The **knee joint** is between the femur and the tibia, with the patella riding in front. It is a modified hinge — primarily flexion and extension, with a small amount of rotation when the knee is flexed. Stability comes from *cruciate ligaments* (ACL and PCL) inside the joint, *collateral ligaments* (MCL and LCL) outside, and two C-shaped *menisci* (fibrocartilage pads) between the femur and tibia.

The **leg** (knee to ankle) has two bones:



Tibia

Medial, larger, weight-bearing. Articulates with femur above and talus below. Easily palpable along its entire anterior border (the shin).



Fibula

Lateral, slender, non-weight-bearing. Provides attachment for muscles and stabilizes the ankle. The lateral malleolus is its distal end.



Patella

The largest sesamoid bone in the body. Embedded in the quadriceps tendon. Improves the mechanical advantage of knee extension.

Twenty-six bones, three arches, built for two jobs

The foot has 26 bones per side: 7 *tarsals* (hindfoot and midfoot), 5 *metatarsals*, 14 *phalanges*. The foot does two jobs: it bears weight and absorbs impact (a rigid structure), and it propels the body forward in walking (a flexible structure). Anatomy that combines those two requires three arches.

Talus

Sits between the tibia/fibula above and the calcaneus below. Receives body weight from the leg and distributes it.

Calcaneus

Heel bone. Largest tarsal. Attachment for the Achilles tendon.

Navicular, cuboid, three cuneiforms

Form the midfoot.

Three arches

Medial longitudinal (highest), *lateral longitudinal*, and *transverse*. Maintained by bone shape, ligaments, and plantar muscles.

DRAW ZONE 1

 -- Sketch the right pelvis and femur. Label the three bones of the os coxae, the acetabulum, the femoral head, neck, greater and lesser trochanters. Mark Eleanor's fracture line and the retinacular blood supply.

Use Fine for the small retinacular arteries that wrap around the neck. These are what got torn.



DRAW ZONE 2 -- Sketch the entire lower limb in anatomical position. Label all bones from the pelvic girdle to the foot. Note the three foot arches.

Star the femoral neck, the medial and lateral malleoli (tibia and fibula), and the calcaneus.

RETURNING TO ELEANOR

Why her leg sits the way it does, and what surgery has to decide

Eleanor's shortened, externally rotated leg is mechanical, predictable, and tells the team the fracture is displaced before X-ray confirms it.

"Why the leg is shortened"

The fracture allows the femoral shaft to ride up relative to the head/acetabulum, pulled superiorly by the hip flexors and quadriceps. The functional length of the limb shortens by the amount of override.

"Why it is externally rotated"

With the neck disrupted, the strong external rotators of the hip (piriformis, obturator internus, quadratus femoris) pull the shaft into external rotation unopposed. The foot points outward.

"Why she could not lift the leg"

Attempting hip flexion fires iliopsoas, which inserts on the lesser trochanter. Without continuity through the femoral neck, the contraction pulls the shaft against the fracture rather than lifting the limb.

"Why blood supply is the big question"

The femoral head is fed by retinacular arteries that wrap the neck. A displaced fracture tears them. If the head loses too much supply, it dies (avascular necrosis), even if the bone heals.

"Hemiarthroplasty vs ORIF"

In an 84-year-old with a displaced femoral neck fracture, surgeons usually **replace** the head with a prosthesis (hemiarthroplasty or total hip). Trying to fix and preserve the head risks AVN, non-union, and a second operation. In younger patients with non-displaced fractures, ORIF preserves the native head when possible.

"Why she needs surgery soon"

Delay increases mortality. The standard is to operate within 24 to 48 hours when medically possible. Bed rest while waiting carries its own risks: pneumonia, DVT, pressure ulcers, delirium.

Pelvis, femur, neck, blood supply. Eleanor's case is the lower extremity boiled down to the most dangerous fracture in the elderly. When you go back to the prework, the diagram of the femoral neck

blood supply will land differently. The arteries you almost skipped are the ones the surgeon is thinking about.